

API, Authentication, Prompting

APIs – The Connectors of Modern Software

An **API (Application Programming Interface)** is the invisible bridge that lets software systems talk to each other. It defines how requests are made, how responses are structured, and ensures smooth communication without exposing internal complexity.

Think of an API as a restaurant waiter: you place an order, the waiter delivers it to the kitchen, and brings back your food. You don't need to know how the kitchen works — just like clients don't need to know server internals.

Key Traits

- **Contract:** Defines inputs, outputs, and behavior.
- **Abstraction:** Hides complexity.
- **Reusability:** One API can serve many clients.
- **Interoperability:** Bridges different languages/platforms.
- **Scalability:** Backend can evolve without breaking clients.

Common Types

- **REST:** HTTP verbs (GET, POST, PUT, DELETE).
- **SOAP:** XML-based, strict contracts.
- **GraphQL:** Clients specify exact data needed.
- **gRPC:** High-performance, binary protocol.
- **WebSockets:** Real-time, two-way communication.

Local Variables – Scoped Data in Action

A **local variable** is declared inside a function or block and exists only within that scope. It's created when the scope begins and destroyed when it ends.

Authentication – Proving Identity

Authentication ensures that only the right users or systems gain access. It answers the question: **"Who are you?"**

Major Types

1. **Basic Authentication:** Username/password sent with each request. Simple but insecure.
2. **API Key Authentication:** Unique key identifies client. Easy but must be kept secret.
3. **Token-Based (JWT/Bearer):** Server issues token after login; client sends it with requests. Stateless and widely used.

4. **Session-Based:** Server stores session; client sends cookie. Common in traditional web apps.
5. **Multi-Factor (MFA):** Combines password + OTP + biometrics. Strongest form.

Error Handling Codes

- **400 Bad Request:** Malformed request.
- **401 Unauthorized:** Invalid/missing credentials.
- **403 Forbidden:** Valid credentials, no permission.
- **429 Too Many Requests:** Rate limit exceeded.
- **500 Internal Server Error:** Server failure.

Prompting – Guiding AI Systems

Prompting is the art of **telling AI exactly what you want**. Just as APIs define contracts between software, prompts define contracts between humans and AI.

Principles of Effective Prompting

- **Clarity:** Be explicit.
- **Context:** Provide background/examples.
- **Structure:** Use steps or lists.
- **Constraints:** Specify style, tone, or length.
- **Iteration:** Refine based on output.

Summary

- **APIs** connect systems seamlessly.
- **Local variables** keep data scoped, safe, and efficient.
- **Authentication** secures access by verifying identity.
- **Prompting** guides AI systems to produce useful results.